Cloud Computing

Exercises - 1 Total points: 10

---

1. Install Oracle VM VirtualBox. VirtualBox is a general-purpose full virtualization software for x86_64 hardware. Download from https://www.virtualbox.org/wiki/Downloads
   a. You can follow these instructions, if your host operating system is Ubuntu https://phoenixnap.com/kb/install-virtualbox-on-ubuntu
   b. You can follow these instructions, if your host operating system is Windows https://www.geeksforgeeks.org/how-to-install-virtualbox-on-windows/

   Once installed, VirtualBox allows users to create and manage virtual machines. The software lets users run multiple guest operating systems concurrently on the host machine.
   c. Create your Virtual Machine (VM) by downloading a guest OS image from https://www.osboxes.org/virtualbox-images/
      Install an Ubuntu virtual machine with VirtualBox. Install a different flavour of Linux, if your host OS is Ubuntu, if you want.
   d. Login to the VM and you can try some basic Linux commands. (2 points)
      Note: Take a screenshot of the terminal of the guest OS from the host OS

2. Connect to one of the SCIS lab servers using SSH from your virtual machine. Ask the TA for the IP of the server. It will be of the form 10.x.x.x
   a. Connect to the shell using the ssh command with your username and password.
      Note: study ssh command in detail
      There is another way to connect to the server using beyond password-based authentication.
   b. Log Into SSH with Keys
      Key-based authentication works by creating a pair of keys: a private key and a public key.
      The private key is located on the client machine and is secured and kept secret.
      The public key can be given to anyone or placed on any server you wish to access.
      When you attempt to connect using a key-pair, the server will use the public key to create a message for the client computer that can only be read with the private key.
      The client computer then sends the appropriate response back to the server and the server will know that the client is legitimate.
      This process is performed automatically after you configure your keys.
      i. To create your public and private keys
         `ssh-keygen -t rsa`
         The file name.pub (satish.pub) is the public key and the second one is the private key.
      ii. Now copy the public key to the remote shell you created
         `ssh-copy-id -i satish.pub snsrirama@IP`
      iii. Now you can login to the server without having to type the password.
         `ssh -i satish snsrirama@IP`

         If you get an error, check that the path to the keyfile is correct and that it has correct rights (`chmod 400 <filename>`)

c. Study the `scp` command in detail and try to copy your studentid.txt file from local machine to CC/Exercise1 folder on your remote shell. (2 points)
Note: Use keys to connect to the shell. Not your password.
Show the file on the shell using tools such as FileZilla/WinSCP.

3. Install Apache server and PHP on the virtual machine
   i. First of all, ensure everything is up to date on your machine
   ```
   sudo apt update
   sudo apt upgrade
   ```
   ii. Install Apache using apt:
   ```
   sudo apt install apache2
   ```
   iii. Confirm that Apache is now running
   ```
   sudo systemctl status apache2
   ```
   iv. Once installed, test by accessing your server's IP in your browser:
   `http://localhost/` or `http://127.0.0.1/`
   v. PHP 7.4 is the latest available right now. So, install that along with some regularly used modules
   ```
   sudo apt install php7.4 php7.4-mysql php-common php7.4-cli
   php7.4-json php7.4-common php7.4-opcache libapache2-mod-
   php7.4
   ```
   vi. Check the installation and version
   ```
   php --version
   ```
   vii. Restart Apache for the changes to take effect:
   ```
   sudo systemctl restart apache2
   ```
   viii. Create a phpinfo.php test page:
   ```
   echo '<?php phpinfo(); ?>' | sudo tee -a
   /var/www/html/phpinfo.php > /dev/null
   ```
   ix. Test everything is working by accessing the following in your browser
   `http://localhost/phpinfo.php`
   You should see a PHP Version 7.4.3 page listing all of your PHP options. If you don't or it tries to download a file, double-check that all of the above steps have completed without error. (2 points)
   Note: Take a screenshot of the page. Customize the page so that you prove you made some modifications to the main page.

4. System administrators who are responsible for web servers often want to monitor these servers and check that they are running correctly. As a HTTP server uses TCP on port 80, the simplest solution is to open a TCP connection on port 80 and check that the TCP connection is accepted by the remote host. However, as HTTP is an ASCII-based protocol, it is also very easy to write a small script that downloads a web page on the server and compares its content with the expected one.
   a. First, use telnet to verify that a web server is running on host https://scislearn2.uohyd.ac.in/moodle/ (1 point)

   Instead of using telnet on port 80, it is also possible to use a command-line tool such as curl Use curl with the --trace-ascii tracefile option to store in tracefile all the information exchanged by curl when accessing the server.

   b. What is the version of HTTP used by curl? (1 point)

c. Please explain the different headers placed by curl in the request? (1 point)
d. Please explain the different headers found in the response? (1 point)

Note: Study curl in detail

Follow these guidelines while submitting the exercises

1. Generate a PDF file with answers/screenshots with name being your studentid_EX1
2. Put a pdf file with the answers for the questions
    a. For programming tasks show screenshots of the program output for the correct and wrong inputs
    b. For commands tried, provide the screenshots
3. Do not copy code/answers from internet. If you copy, please leave a comment with the URL on the file
4. Plagiarism will not be tolerated and dealt through the formal channels of the university